

APSC 131 – Chemistry of Materials

Lecture 03 – Chemistry Fundamentals

Learning Plan

Keywords

- Valence, Lewis structures, mole
- Reaction stoichiometry, yield and atom economy

Resources

- See curated OnQ video list

Learning Outcomes

- Calculate reagent masses and atom economy for a balanced chemical reaction

Elements, Compounds and Mixtures

A chemical ***element*** is the building-blocks of matter, defined a pure substance of atoms with the same number of protons.

A chemical ***compound*** is any substance composed of identical molecules consisting of atoms of two or more chemical elements.

A chemical ***mixture*** is a blend whose components to not form covalent bonds:

Homogeneous mixtures are single-phase liquid solutions and solid alloys with evenly distributed elements or compounds. Examples include saltwater, brass, and air.

Heterogeneous mixtures are multi-phase systems with distinguishable boundaries, and can be comprised of solids, liquids and/or gases. Examples include suspensions such as muddy water, milk colloids of fat particles in water, and ores such as granite.

Composition of Mixtures

Mass fraction (dimensionless): Used in production facilities where mass measurements are convenient, the mass fraction of component i in a mixture is:

$$w_i = \frac{m_i}{\sum m_i}; \quad \sum w_i = 1$$

Mole fraction (dimensionless): Used widely in phase and reaction equilibrium calculations, the mole fraction of component i in a mixture is:

$$x_i = \frac{n_i}{\sum n_i}; \quad \sum x_i = 1$$

Molality (mol/g): Rarely used, the molality of a component of mixture = $n_i / \sum m_i$

Molar concentration (mol/litre): The composition of homogenous liquid solutions are typically expressed as moles of compound per unit volume, called the molarity = $n_i / Volume$

Formal concentration (mol/litre): If the dissolved compound is reactive (i.e. NaCl or acetic acid in water) we report its formality, since NaCl dissociates into its ions and acetic acid dissociates to some degree. Formality = $n_i / Volume$

Lewis Symbols and Structures

A Lewis symbol consists of a symbol to represent the nucleus and *core* (inner shell) electrons of an atom, together with dots placed around the symbol to represent the *valence* (outer shell) electrons.

- The Lewis symbol for carbon, which has the electron configuration $[\text{He}]2s^22p^2$, is $\cdot\dot{\text{C}}\cdot$.

A Lewis structure is a combination of Lewis symbols that represents the formal distribution of electrons in covalent and ionic bonds.

Group	1	2	3	4	5	6	7	8
Number of valence e ⁻	1	2	3	4	5	6	7	8
Valence	1	2	3	4	3	2	1	0
Lewis dot valence state	Li ·	· Be ·	· B ·	· C ·	: N ·	: O :	: F :	: Ne :

Draw the molecular formula and Lewis structures for diethyl ether and lithium fluoride.

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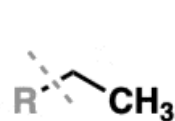
Lewis Structures of Common Organic Functional Groups

It is not necessary to draw protons or label the carbons of a Lewis structure, but feel free to do so if it helps with your analysis.

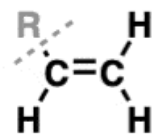
Drawing the lone pairs of heteroatoms (O, N, S, halide) is recommended, at least while you are familiarizing yourself with functional group chemistry.

Functional Groups - The Main Players

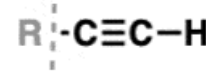
R is a carbon substituent



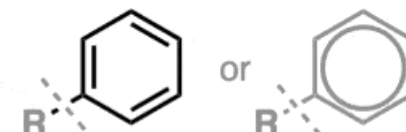
alkane
(-alkyl)



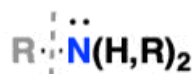
alkene
(-alkenyl)



alkyne
(-alkynyl)



benzene ring
(-phenyl)



amine
(-amino)



alcohol
(-hydroxy)



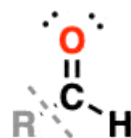
ether
(-alkoxy)



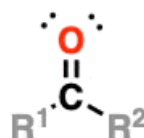
alkyl halide



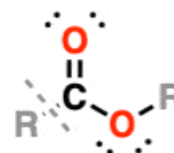
thiol



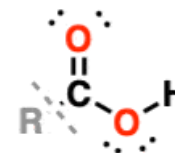
aldehyde



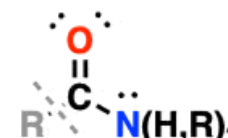
ketone



ester

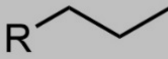
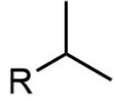
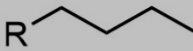
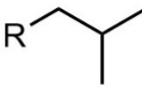
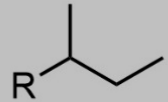
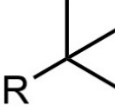
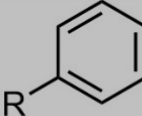


carboxylic
acid



amide

Lewis Structures of Common Organic Functional Groups

functional group	name	abbreviation(s)
	methyl	Me
	ethyl	Et
	propyl	Pr (or <i>n</i> Pr or <i>n</i> -Pr or ^{<i>n</i>} Pr)
	<i>iso</i> -propyl	<i>i</i> Pr (or <i>i</i> -Pr or ^{<i>i</i>} Pr)
	butyl	Bu (or <i>n</i> Bu or <i>n</i> -Bu or ^{<i>n</i>} Bu)
	<i>iso</i> -butyl	<i>i</i> Bu (or <i>i</i> -Bu or ^{<i>i</i>} Bu)
	<i>sec</i> -butyl	<i>s</i> Bu (or <i>s</i> -Bu or ^{<i>s</i>} Bu)
	<i>tert</i> -butyl	<i>t</i> Bu (or <i>t</i> -Bu or ^{<i>t</i>} Bu)
	phenyl	Ph

The Mole and the Molar Mass of a pure substance

Mass is a physical measurement of the *amount of matter in a system*, but it is not of direct significance in chemistry.

- Chemical properties and chemical reactions are derived from molecular interactions
- One gram of Substance A contains a different number of molecules than Substance B

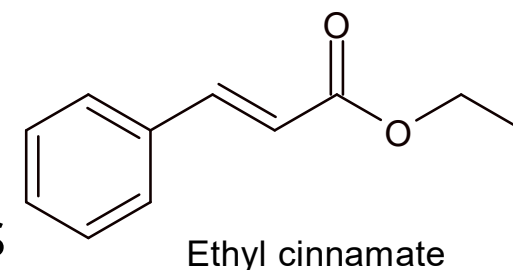
The mole is a calculated measure of the *amount of a substance in the system*.

- $1 \text{ mol} = 6.022 \times 10^{23}$ elementary atoms/molecules = Avogadro's number (N_A)

Converting the mass of an element / compound to moles requires a molar mass calculation based on atomic mass units:

$$1 \text{ amu} = 1 \text{ g/mol}$$

Calculate the number of moles and the number of molecules of a system comprised of 1 kg of ethyl cinnamate ($\text{C}_{11}\text{H}_{12}\text{O}_2$):



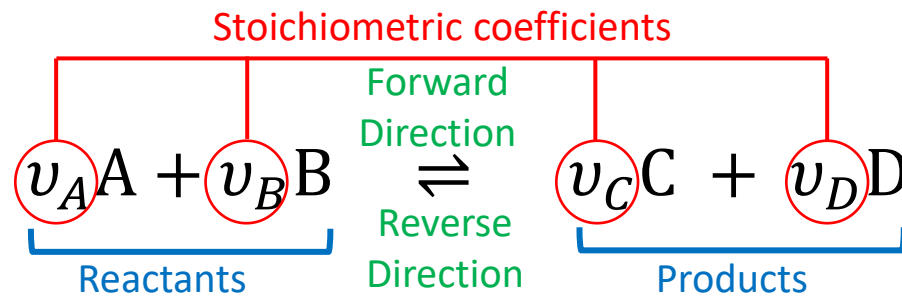
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Reaction Stoichiometry and Balancing

Chemical reactions do not alter the atomic number of elements in a system, only valence electron populations and/or bonds between atoms.

- This excludes **nuclear** chemistry such as fission and fusion, where the composition of elements changes.

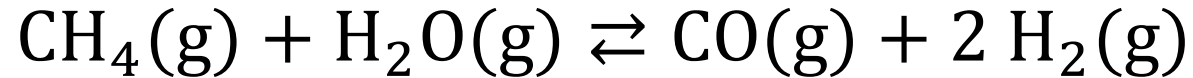
Consider a system in which compounds A and B are equilibrium with C and D:



- A reaction equation must be **balanced** in terms of the number of atoms of each element on each side.
- v_A , v_B , v_C , and v_D are molar stoichiometric coefficients.

Reaction Stoichiometry and Balancing

The water-gas shift reaction is used industrially to produce hydrogen gas from methane according to the following equilibrium equation,

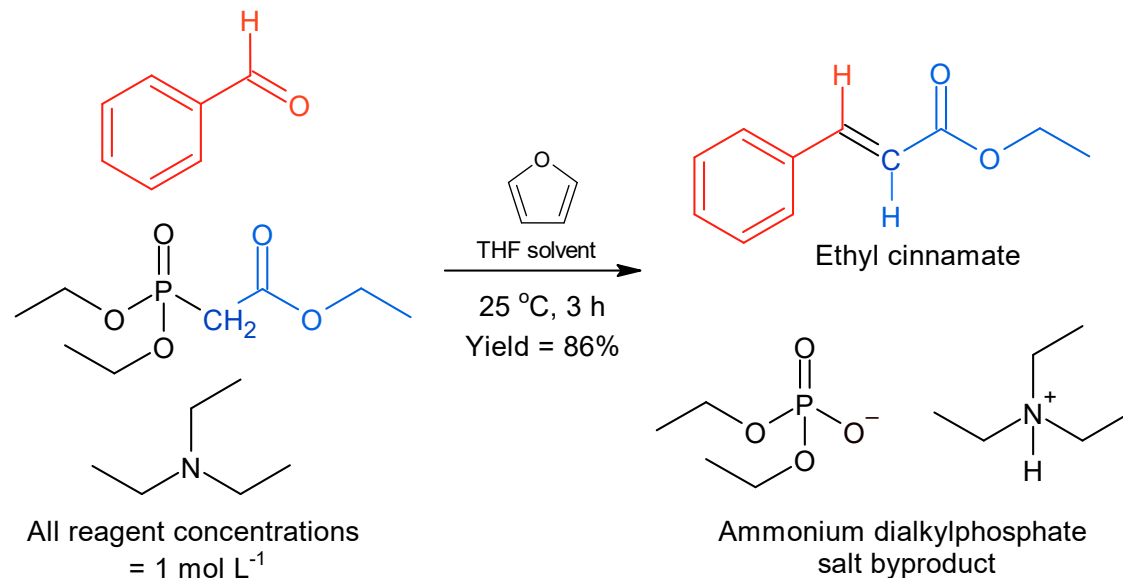


- Assign stoichiometric coefficients for each component.
- Determine whether equation is atomically balanced.
- Calculate the change in the **total number of moles** in the system if 1 mole of methane is consumed by this reaction.
- Calculate the change in the **total mass of the system** if 1 mole of methane is consumed by this reaction.
- Account for why total mass is conserved, but total moles is not.

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Reaction Stoichiometry and Balancing

Ethyl cinnamate is a food flavor additive that is used in many fragrance compounds.



What are the stoichiometric coefficients of each reactant and product?

Is this chemical equation balanced?

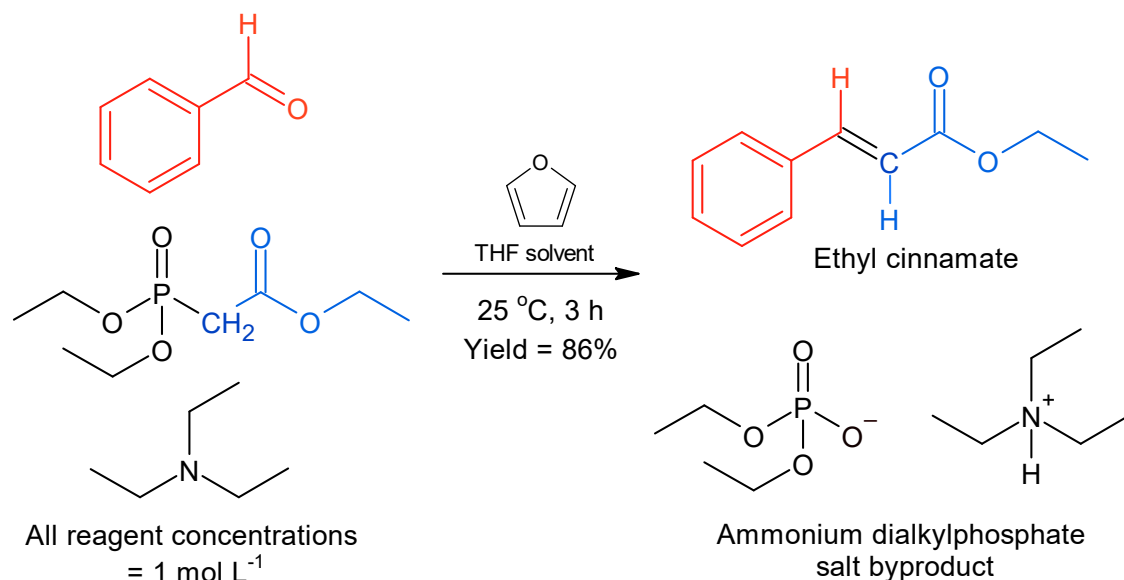
Note that reaction engineers are concerned not only with the product yield, but also the structure and amounts of **byproducts** that must be handled responsibly.

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Performance Metrics: Yield

The most widely reported reaction performance metric provides the mass of the target molecule divided by the maximum theoretical mass based on the limiting reagent.

$$\text{Yield} = \frac{\text{actual amount of product}}{\text{theoretical amount of product based on limiting reagent}} = 86\%$$

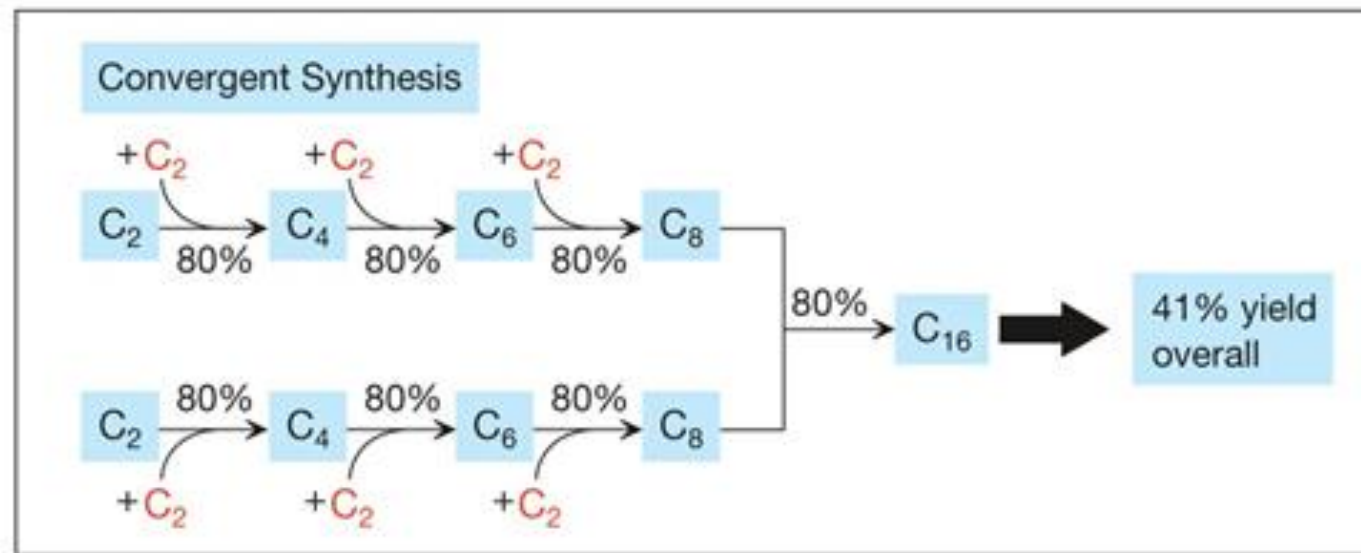
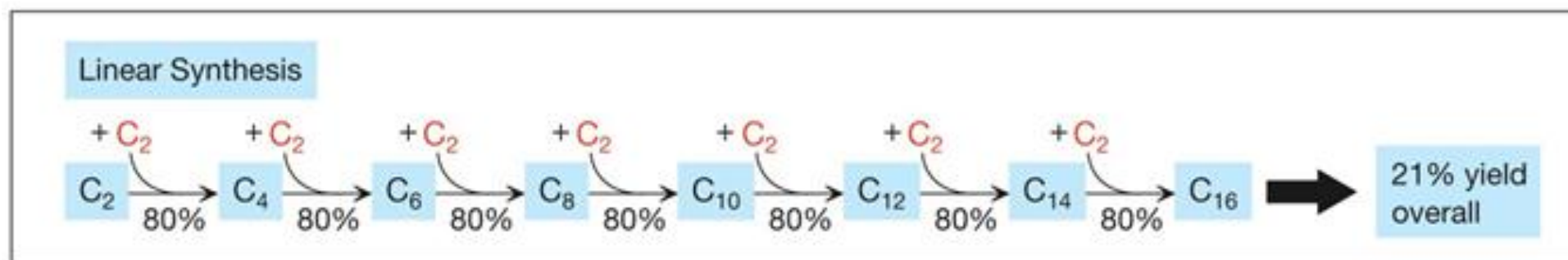


If asked to produce 1 kg of ethyl cinnamate by this process, calculate the composition of the reaction mixture:

- Mass of aldehyde
- Mass of phosphonate
- Mass of triethyl amine
- Approximate volume of THF solvent

Convergent Syntheses: Maximizing Reaction Yield

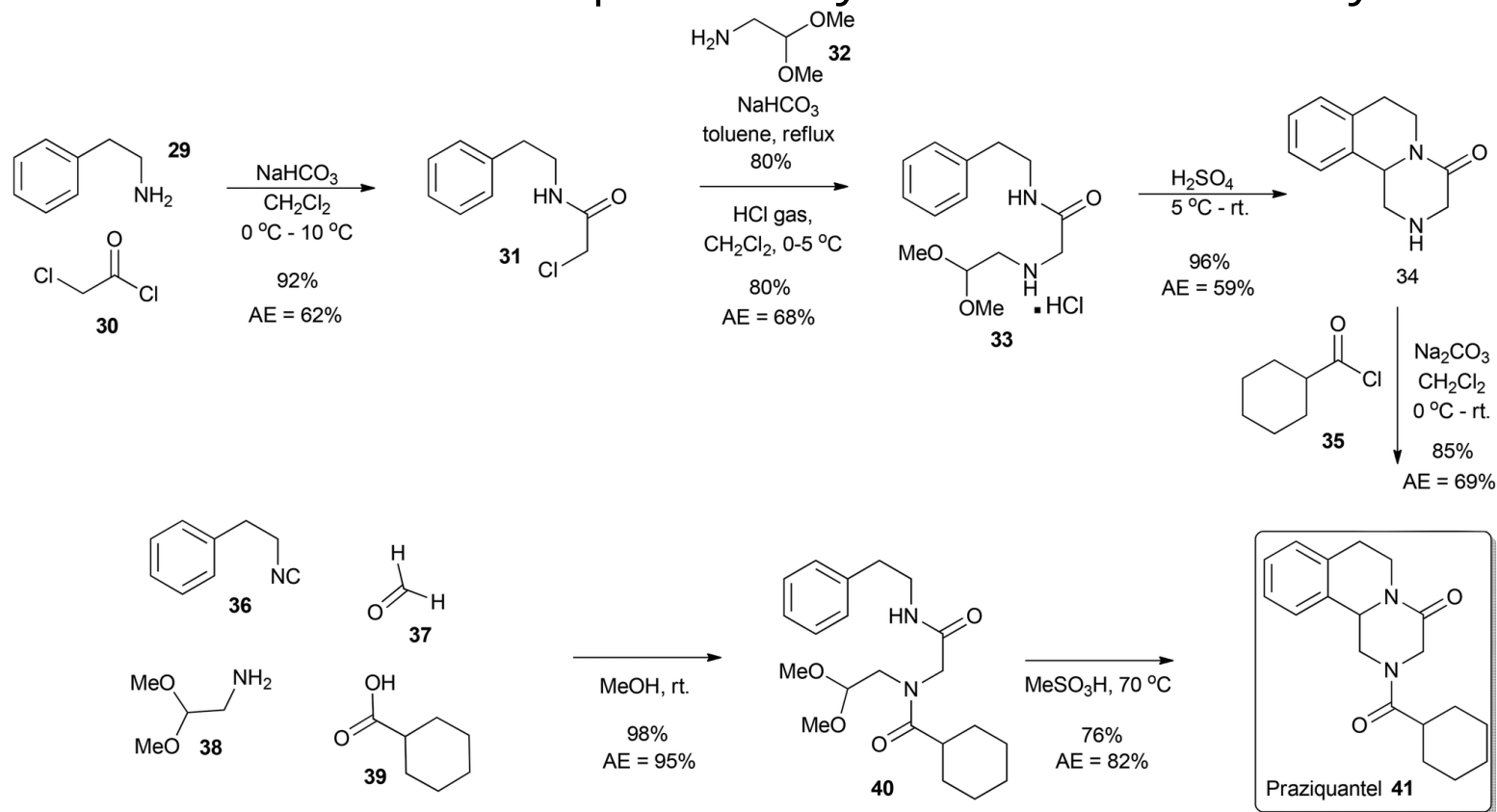
Linear syntheses involving long reaction sequences are less efficient than convergent syntheses that construct a target molecule from larger reagents.



Both syntheses involve 7 reactions of 80% yield, yet the convergent approach is twice as efficient overall.

Convergent Syntheses: Maximizing Reaction Yield

Praziquantel is on the WHO's list of essential medicines, used to treat schistosomiasis (snail fever) in the developing world. It is synthesized commercially in five steps by a robust, high yielding process that does not involve particularly hazardous chemistry.

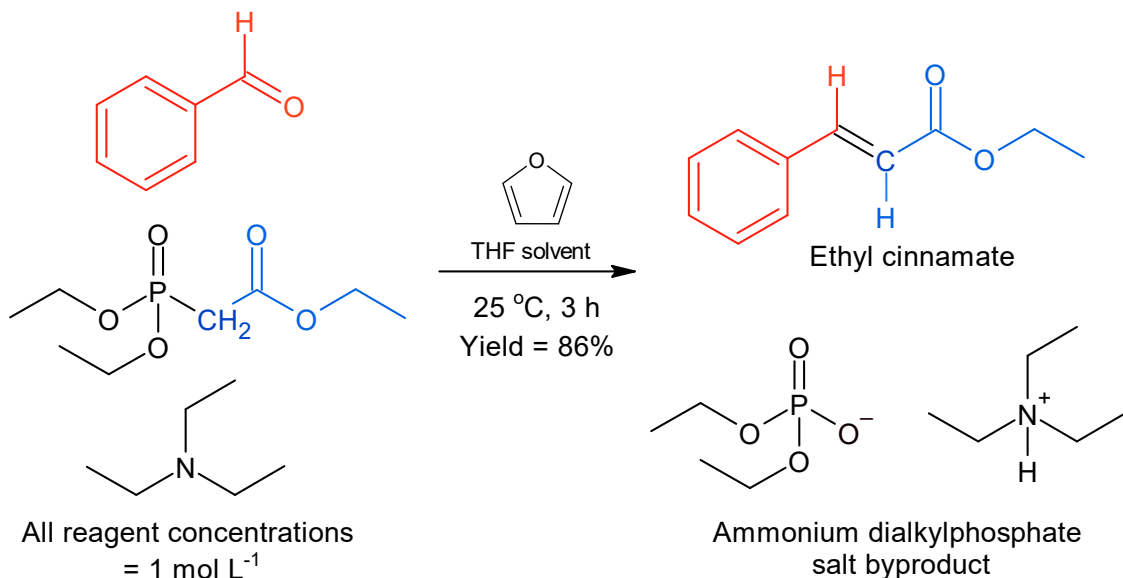


However, the two-step convergent synthesis uses the same cyclization strategy, but a four-component reaction called the Ugi reaction.

Performance Metrics: Atom Economy

Atom economy is a Green Chemistry metric of how much of the reagents ends up in the target molecule, and how much ends up as reaction byproducts.

- Ideally, every atom within the reagents is found in the synthesized product.
- Note that this calculation does **not** involve the reaction yield (assumes that the reaction can achieve 100% of the theoretical mass).



$$\text{Atom Economy} = \frac{\text{theoretical mass of product}}{\text{mass of all reactants}} = 41\%$$

Almost 60% of the material charged to the reaction will be lost to byproducts (even assuming a 100% yield)!

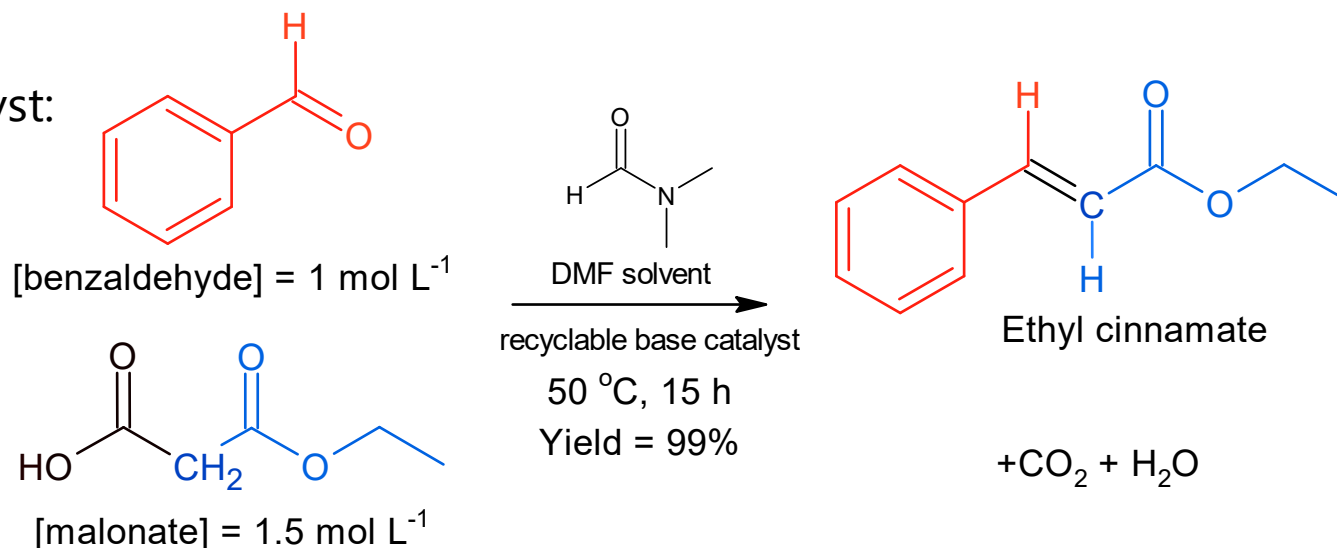
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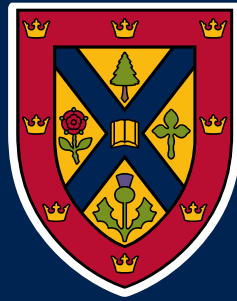
Knowledge Development Questions

Performance of Ethyl Cinnamate Synthesis

Consider the alternate synthesis of ethyl cinnamate that uses solid, recyclable catalyst:

- Determine** the stoichiometric coefficients of all reactants and products.
- Assess** whether the chemical equation is balanced.
- Calculate** the composition of the reaction mixture required to produce 1 kg of ethyl cinnamate by this synthesis:
 - Mass of aldehyde
 - Mass of malonate
 - Approximate volume of DMF solvent
- Estimate** the atom economy of this synthesis and compare its efficiency with that of the lecture example.





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